

DAT246 Practice Exam – Questions & Answers

(Bayesian Data Analysis & Empirical Software Engineering)

Part I – Exam Questions (45–60 min total)

- 1■■■ Describe the Bayesian data analysis workflow and its main stages.
 - 2■■■ Explain three principled Bayesian methods for avoiding overfitting and provide one example for each.
 - 3■■■ Compare experiments and observational studies in terms of validity threats. What tension exists between them?
 - 4■■■ Define information entropy and Kullback–Leibler divergence. How are they used in model comparison?
 - 5■■■ What is the maximum entropy principle and how does it justify the choice of likelihoods in generalized linear models (GLMs)?
 - 6■■■ Explain the basic idea of Markov Chain Monte Carlo (MCMC) sampling and how we check convergence and mixing.
 - 7■■■ Define complete pooling, no pooling, and partial pooling. Why is partial pooling usually preferred?
 - 8■■■ When should we use mixture models or zero-inflated models? How do they differ from simple Poisson or Binomial models?
 - 9■■■ Explain how covariance is modeled in multilevel models, including the role of the LKJ(η) prior. What does η control?
- Applied question: You observe bug counts from multiple software teams, many with zeros and strong differences between teams. Design a suitable Bayesian model, name its components, and justify your modeling choices.

Part II – Model Answers & Key Concepts

1 ■ Bayesian workflow: define causal DAG → choose likelihood $p(y|\theta)$ and priors $p(\theta)$ → run prior predictive check → fit model via MCMC → check diagnostics ($R^2 \approx 1$, $n_{\text{eff}} \uparrow$, no divergents) → posterior predictive check → compare models (WAIC/LOO) → interpret posterior. (■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■).

2■■■ (1) Regularizing priors: constrain parameters (e.g., $\text{Normal}(0,1)$). (2) Information criteria: WAIC/PSIS-LOO assess predictive accuracy. (3) Multilevel modeling: partial pooling shares information. Together they reduce variance and overfitting.

3. Experiments → high internal validity (high) but low external (low); Observations → high external but low internal (low). Tension = control vs realism tradeoff.

4 ■ Entropy $H(p) = -\sum p \log p$ measures uncertainty. KL divergence $D_{\text{KL}}(p||q) = \sum p \log(p/q)$ = information loss when $q \approx p$. Criteria WAIC/LOO approximate $-KL$ to evaluate predictive accuracy.

5 ■ Maximum entropy principle: choose the flattest distribution consistent with known constraints → Normal (μ, σ) for mean+variance, Binomial for binary, Poisson for counts. GLMs link linear predictors to these likelihoods using link functions (logit/log).

6 ■ ■ ■ MCMC builds a Markov chain whose stationary distribution $\approx$ posterior $p(\theta|y)$. Algorithms: Metropolis (Propose–Accept–Repeat), HMC (uses gradients). Diagnostics: $R \approx 1$, large n_{eff} , good trace mixing, no divergent transitions.

7 ■■ Complete pooling: one global parameter; No pooling: separate for each group; Partial pooling: group parameters $\sim \text{Normal}(\alpha, \sigma)$. Partial pooling balances bias & variance $\rightarrow$ better generalization (■■■■).

8 ■ Mixtures model overdispersion (variance > mean): Beta-Binomial, Neg-Binomial. Zero-inflated models add an extra zero process (ZIP). Mixtures → heterogeneity; ZI → excess zeros.

9 ■ In MLMs, varying intercepts/slopes $\sim \text{MVN}([\alpha \quad \beta], \Sigma)$. Σ has variances σ_{α^2} , σ_{β^2} and correlation ρ . LKJ(η) prior regularizes correlation: $\eta = 1$ uniform, $\eta > 1$ shrinks toward 0.

- Use a hierarchical Zero-Inflated Negative Binomial (ZINB):

$$y_{ij} \sim \text{ZINB}(\lambda_{ij}, \phi, \pi); \log \lambda_{ij} = \alpha_{\text{team}[j]} + \beta x_{ij};$$

$\alpha_{\text{team}} \sim \text{Normal}(\alpha_{\text{team}}, \sigma_{\alpha_{\text{team}}})$. $\rightarrow$ Handles overdispersion + excess zeros + team hierarchy.